

Multimodal Chest X-ray Abnormality Prediction with Cross-Modal Attention Analysis

Haoran Ding

Abstract

Chest X-ray interpretation is a fundamental task in clinical radiology, yet accurate automated abnormality detection remains challenging due to severe class imbalance, weak supervision, and limited interpretability. In this work, we develop a multimodal framework that jointly leverages chest X-ray images and radiology reports for abnormality prediction on the Indiana University Chest X-ray dataset. The proposed approach integrates a vision encoder with ClinicalBERT-based text representations through cross-modal attention.

Experiments were conducted on five thoracic abnormalities: effusion, cardiomegaly, edema, pneumonia, and pneumothorax. Initial evaluation revealed that the model achieved strong ranking performance despite near-zero predictions under the default threshold of 0.5. We therefore performed label-specific threshold optimization, which substantially improved F1 scores and revealed that calibration and operating-point selection are critical for highly imbalanced clinical datasets.

The best-performing model achieved AUROC scores of 0.847 for cardiomegaly and 0.881 for edema. Cross-attention visualization further demonstrated partial alignment between textual clinical concepts and anatomically relevant thoracic regions. However, several failure cases highlighted the challenges introduced by noisy report-derived supervision and severe class imbalance.

1 Introduction

Chest X-ray imaging is one of the most widely used diagnostic modalities in modern medicine. Automated chest X-ray interpretation has therefore become an important research direction in medical artificial intelligence. Recent advances in deep learning and multimodal learning have enabled models to jointly process medical images and associated radiology reports, providing opportunities for improved clinical reasoning and interpretability.

Despite substantial progress, several challenges remain. First, many chest X-ray datasets exhibit extreme class imbalance, where positive abnormalities represent only a small fraction of the dataset. Second, most publicly available labels are weakly supervised and automatically extracted from free-text radiology reports. As a result, labels may contain ambiguity, uncertainty, or negated findings. Third, model interpretability remains difficult, especially in multimodal architectures where cross-modal interactions are often poorly understood.

In this study, we investigate multimodal abnormality prediction using the Indiana University Chest X-ray dataset. Our framework combines image embeddings with ClinicalBERT text representations through cross-attention. We evaluate the model on five thoracic abnormalities and analyze both quantitative performance and qualitative attention behavior.

Our contributions are summarized as follows:

- We develop a multimodal chest X-ray abnormality prediction framework integrating image and text information.

- We demonstrate that label-specific threshold optimization substantially improves performance under highly imbalanced medical settings.
- We provide detailed ROC, precision-recall, and threshold calibration analyses.
- We visualize cross-modal attention maps and discuss both successful and failure cases under weak supervision.

2 Dataset

2.1 Indiana University Chest X-ray Dataset

Experiments were conducted using the Indiana University Chest X-ray collection. The dataset contains frontal chest radiographs paired with associated radiology reports consisting of findings and impression sections. Five thoracic abnormalities were selected for evaluation:

- Effusion
- Cardiomegaly
- Edema
- Pneumonia
- Pneumothorax

The dataset exhibits substantial class imbalance. Figure 1 summarizes the positive sample distribution.

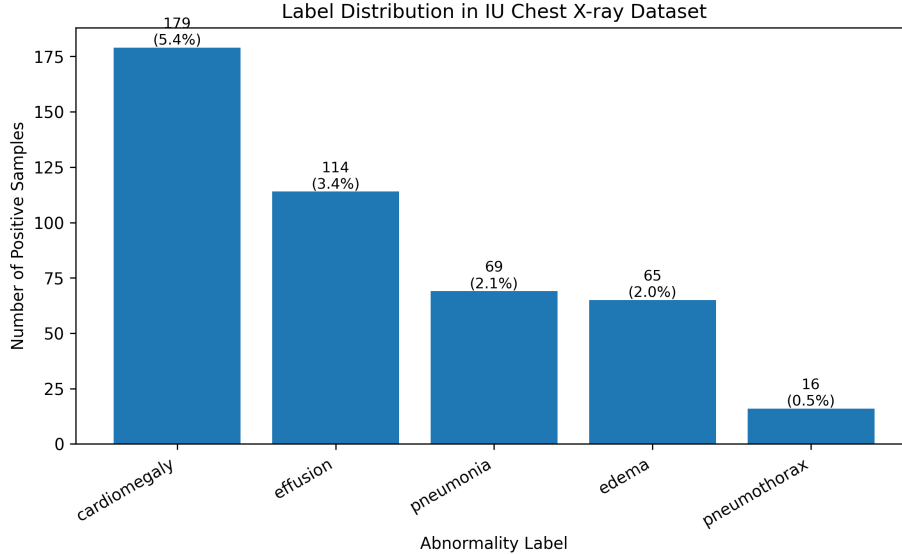


Figure 1: Positive sample distribution for the five thoracic abnormalities in the Indiana University Chest X-ray dataset.

3 Methodology

3.1 Overall Framework

The proposed framework jointly models chest X-ray images and radiology report text through a multimodal vision-language architecture. Clinical text tokens query visual image patches through cross-modal attention fusion. The overall pipeline is illustrated in Figure 2.

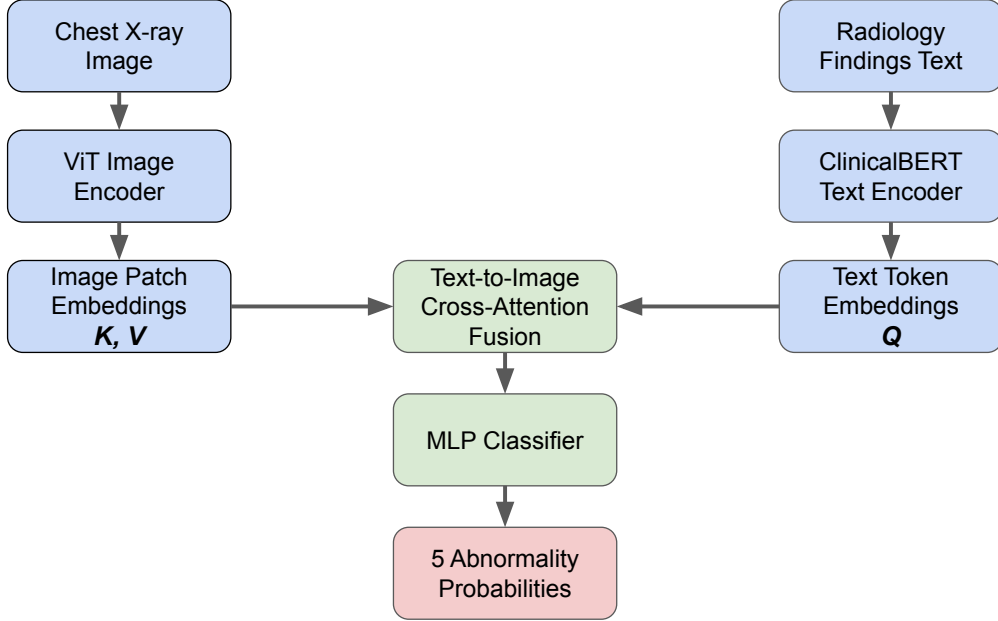


Figure 2: Overview of the multimodal chest X-ray framework.

The framework consists of three major components:

1. A vision encoder that extracts spatial image patch embeddings from chest X-ray images
2. A ClinicalBERT-based text encoder that extracts contextualized clinical language representations
3. A cross-modal attention fusion module that allows textual clinical concepts to attend to image regions

The final multimodal representation is subsequently passed to a multilabel classification head for abnormality prediction.

3.2 Image Encoder

The vision branch processes chest X-ray images using a Vision Transformer (ViT)-style encoder. The input image is divided into spatial patches, which are then projected into dense embedding vectors:

$$X_{img} = \{x_1, x_2, \dots, x_n\}$$

where each embedding corresponds to a local image patch. These patch-level embeddings preserve spatial information and allow the model to associate clinical concepts with localized thoracic regions.

3.3 Text Encoder

Text embeddings were extracted using pretrained ClinicalBERT representations:

$$X_{txt} = \{t_1, t_2, \dots, t_m\}$$

where each token embedding captures contextualized clinical semantics.

3.4 Cross-Modal Attention Fusion

The central component of the framework is the cross-modal attention fusion mechanism. Instead of simply concatenating image and text embeddings, the proposed approach allows clinical text representations to actively attend to spatial image regions. Specifically, text token embeddings are used as queries (Q), while image patch embeddings are used as keys (K) and values (V):

$$Q = X_{txt}, \quad K = X_{img}, \quad V = X_{img}$$

Cross-attention is computed using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

This mechanism enables disease-related clinical concepts such as “cardiomegaly”, “edema”, or “consolidation” to selectively attend to visually relevant thoracic regions. The resulting attention-weighted multimodal representation integrates both semantic clinical context and spatial image information.

3.5 Classification Head

The fused multimodal representation is aggregated and passed through a multilayer perceptron (MLP) classification head. Final abnormality probabilities are generated using sigmoid activation for multilabel prediction:

$$\hat{y} = \sigma(Wz + b)$$

where z denotes the fused multimodal representation.

4 Experimental Setup

4.1 Text Processing

The “Findings” section was used as the textual input to the multimodal framework. Weak labels were derived from the “Impression” section using rule-based regular expression matching.

4.2 Image Processing

Images were resized and normalized before model training to ensure consistent input dimensions for the vision encoder.

4.3 Data Split

Experiments were conducted on the Indiana University Chest X-ray dataset using an 80/20 random train-validation split. Only frontal chest X-ray images were included in the study.

The validation set was used for all quantitative evaluations, including AUROC, AUPRC, precision, recall, F1 score, and threshold calibration analysis.

4.4 Training Configuration

The multimodal framework was implemented in PyTorch. Clinical text inputs were encoded using pretrained Bio.ClinicalBERT representations, while chest X-ray images were processed using a Vision Transformer (ViT)-style image encoder.

Models were trained using binary cross-entropy (BCE) loss for multilabel abnormality prediction. Both unweighted BCE and weighted BCE settings were explored during the ablation study.

The following training configuration was used:

- Optimizer: AdamW
- Batch size: 8
- Maximum text length: 128
- Number of epochs: 5
- Image resolution: 224×224

4.5 Evaluation Metrics

Performance was evaluated using the following metrics:

- Area Under the ROC Curve (AUROC)
- Area Under the Precision-Recall Curve (AUPRC)
- Precision
- Recall
- F1 score

Given the severe class imbalance in the dataset, AUROC and AUPRC were considered primary ranking metrics.

5 Experimental Results

5.1 Main Quantitative Results

Table 1 summarizes the final performance of the best-performing model (unweighted epoch 4).

Table 1: Main evaluation results for the best-performing model.

Label	Pos Count	Pos Ratio	AUROC	AUPRC	Threshold	F1	Recall
Effusion	21	0.032	0.807	0.141	0.04	0.277	0.619
Cardiomegaly	36	0.054	0.847	0.270	0.18	0.324	0.306
Edema	13	0.020	0.881	0.197	0.15	0.300	0.231
Pneumonia	6	0.009	0.855	0.044	0.05	0.083	0.167
Pneumothorax	5	0.008	0.715	0.018	0.50	0.000	0.000

Results indicate that the model achieved strong ranking performance despite severe imbalance. In particular, edema and cardiomegaly achieved AUROC values above 0.84.

5.2 ROC Curve Analysis

Figure 3 presents ROC curves for all abnormalities.

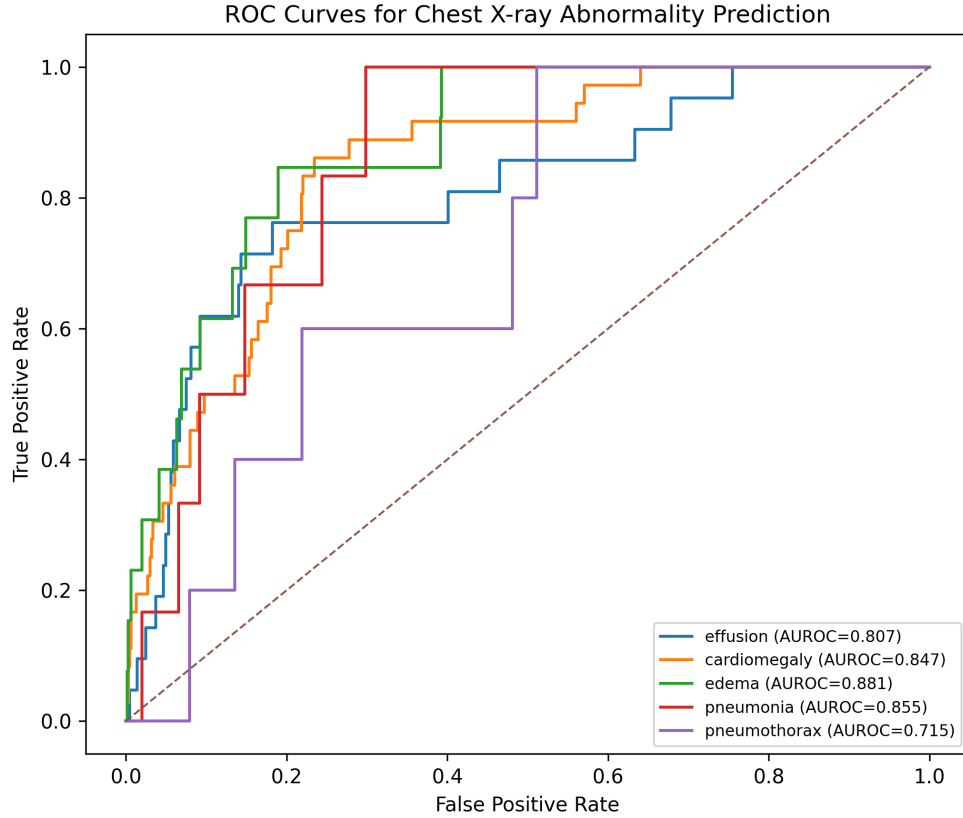


Figure 3: ROC curves for chest X-ray abnormality prediction.

The ROC analysis demonstrates that the model learned clinically meaningful ranking signals even under highly imbalanced settings.

5.3 Precision-Recall Curve Analysis

Figure 4 presents precision-recall curves.

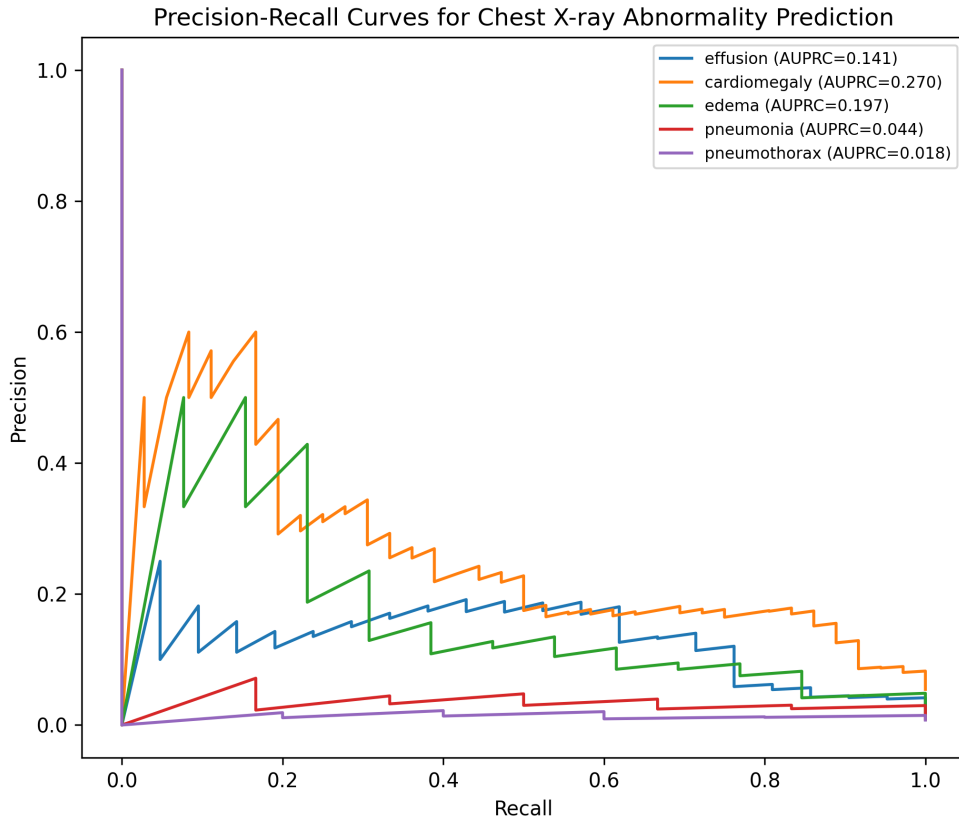


Figure 4: Precision-recall curves for chest X-ray abnormality prediction.

Compared with ROC curves, precision-recall performance was substantially lower due to extreme positive rarity. This behavior is expected in highly imbalanced medical datasets.

5.4 Threshold Calibration Analysis

Initial experiments revealed that the default classification threshold of 0.5 produced near-zero positive predictions for several abnormalities despite relatively strong AUROC performance.

To better analyze operating-point behavior, label-specific threshold optimization was performed on the validation set. Thresholds ranging from 0.01 to 0.50 were evaluated in increments of 0.01:

$$T = \{0.01, 0.02, \dots, 0.50\}$$

For each abnormality, the threshold maximizing F1 score was selected as the optimal operating point.

This threshold calibration procedure reflects common clinical machine learning practice, where deployment operating points are often task-dependent and highly sensitive to class imbalance.

Figure 5 illustrates the relationship between classification threshold and F1 score.

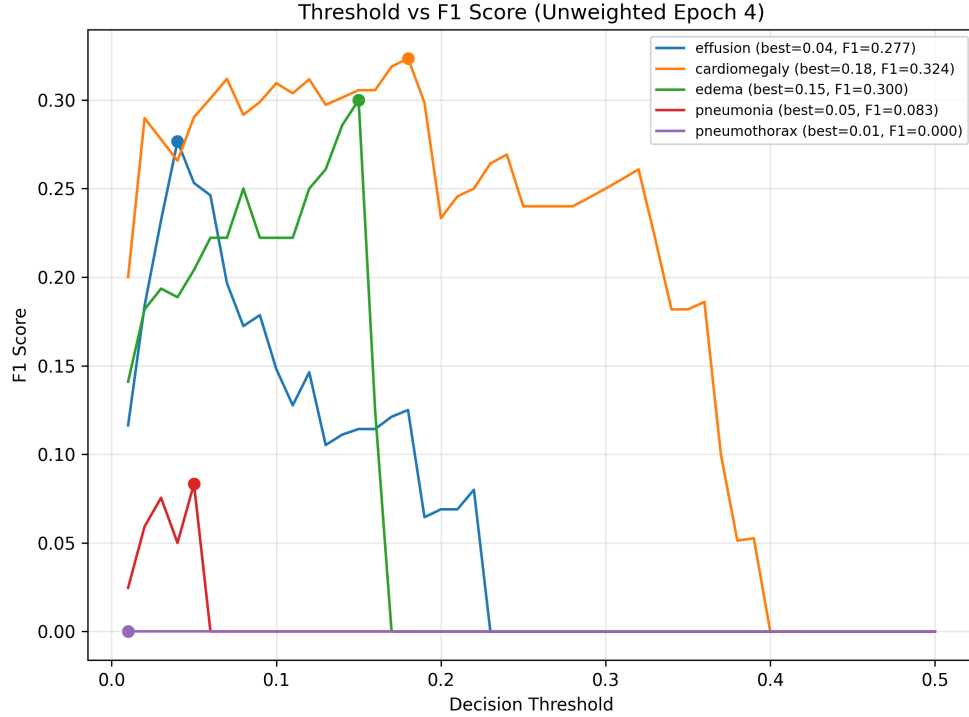


Figure 5: Threshold versus F1 score analysis for the best-performing model.

Several abnormalities achieved optimal performance at very low thresholds (e.g., 0.04 for effusion and 0.05 for pneumonia). This finding highlights that the default threshold of 0.5 is inappropriate for highly imbalanced clinical prediction tasks.

5.5 Ablation Study

Table 2 compares weighted and unweighted training strategies.

Table 2: Ablation analysis across training strategies and epochs.

Setting	Macro AUROC	Macro AUPRC	Macro F1
Unweighted Epoch 1	0.789	0.099	0.197
Unweighted Epoch 2	0.807	0.109	0.173
Unweighted Epoch 3	0.817	0.121	0.211
Unweighted Epoch 4	0.821	0.134	0.197
Unweighted Epoch 5	0.802	0.145	0.204
Weighted Epoch 1	0.764	0.089	0.067
Weighted Epoch 2	0.754	0.073	0.130
Weighted Epoch 3	0.774	0.146	0.158
Weighted Epoch 4	0.796	0.149	0.122
Weighted Epoch 5	0.811	0.134	0.125

Weighted training increased recall but often degraded ranking performance and calibration quality. The unweighted strategy provided a better balance between AUROC, AUPRC, and interpretability.

5.6 Cross-Modal Attention Case Study

To qualitatively investigate multimodal alignment behavior, cross-attention heatmaps were visualized for representative validation samples.

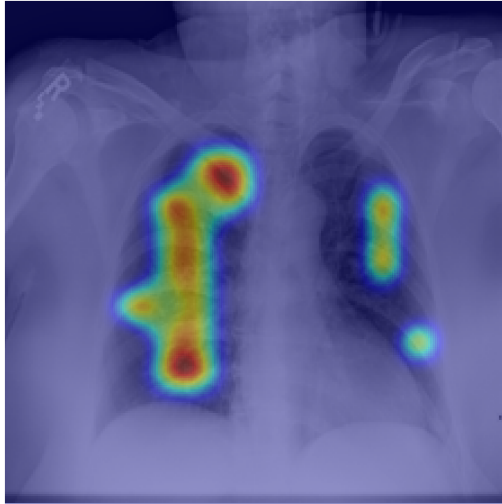


Figure 6: Cross-attention visualization for a pneumothorax case.

Figure 6 shows a pneumothorax example in which the model primarily attends to upper lung boundary regions and pleural margins. Although the localization is not perfectly precise, the attention distribution partially overlaps with anatomically relevant thoracic regions commonly associated with pneumothorax findings.

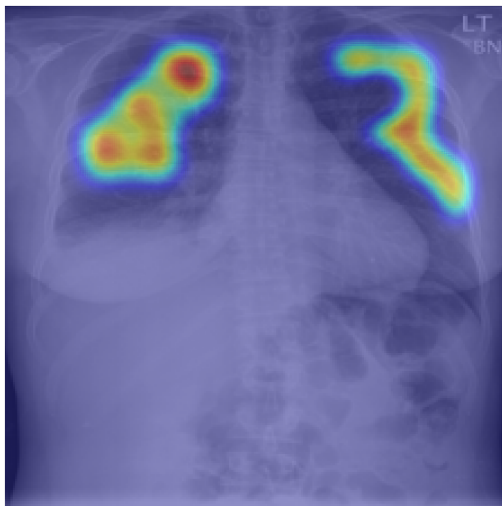


Figure 7: Cross-attention visualization for a pneumonia/consolidation case.

Figure 7 presents a pneumonia/consolidation example. In this case, the model exhibits clustered attention patterns over the right upper lung field rather than isolated random activations. The attention map demonstrates that clinical text tokens can guide the model toward spatially meaningful image regions through cross-modal fusion.

However, attention localization remained imperfect across many samples. Several failure cases revealed diffuse or anatomically implausible attention distributions, likely caused by weak supervision derived from rule-based report labels, severe class imbalance, and the absence of explicit localization annotations.

Overall, these visualizations suggest that the proposed framework learned partial cross-modal grounding behavior despite being trained only with weak image-level supervision.

6 Discussion

Several important observations emerged from this study.

First, the model achieved relatively strong AUROC performance despite severe class imbalance and limited positive samples. This suggests that multimodal learning successfully captured meaningful ranking signals.

Second, threshold optimization proved essential. The default threshold of 0.5 was unsuitable for highly imbalanced chest X-ray prediction tasks. Label-specific operating-point selection substantially improved F1 performance.

Third, weighted training increased recall but often degraded ranking quality and calibration. These findings suggest that preserving probabilistic ranking behavior may be more important than aggressively balancing the loss function.

Fourth, attention visualization revealed both strengths and limitations of multimodal learning. Some cases demonstrated clinically meaningful localization behavior. However, noisy supervision from report-derived labels frequently limited precise grounding.

Finally, several failure cases suggested shortcut learning behavior, where the model relied on positional artifacts, image borders, or acquisition-related features rather than true pathology localization. This phenomenon is common in medical imaging AI and remains an important challenge for future work.

7 Conclusion

This study presented a multimodal chest X-ray abnormality prediction framework integrating radiographic images and radiology reports through cross-modal attention.

Experimental results demonstrated that:

- Unweighted training combined with threshold optimization achieved the best overall balance.
- Strong AUROC performance can coexist with poor default-threshold classification behavior.
- Cross-modal attention partially aligned with clinically meaningful thoracic regions.
- Weak report-derived supervision remains a major challenge for reliable localization.

Future work may include larger-scale datasets, improved report parsing, uncertainty-aware supervision, and clinically grounded attention regularization.